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Nano-layered-structure interface between Sn-Ti alloy and quartz glass for hermetic seals

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ABSTRACT

The interface structure between Sn-Ti alloy and quartz glass was investigated by transmission electron microscope. The results show that the interface is continuous and dense and has a nano-layered-structure with thickness of several tens of nanometer without holes and cracks. The interface is divided into two layers. One is the TiO_2 layer with a thickness of about 2 nm locates near Sn-Ti alloy, and the other is a multiphase structure layer near the quartz glass composed of nanocrystalline $TiSi_2$, Ti_5Si_3 and TiO_2.

1. Introduction

Glass-metal sealing is widely used in the fields of electric vacuum device, light source, electron, solar receiver tube [1], laser and lithium battery. The matching sealing between different kinds of glass and Kovar alloy or stainless steel has been widely studied [1–6]. Quartz glass-metal sealing belongs to a typical unmatched sealing. Quartz glass has excellent high temperature resistance, and its softening temperature reaches 1580 °C. So it is difficult to connect with metal by melting. Additionally, the thermal expansion coefficient of quartz glass is about 5.6×10^{-7}/K, which is about one order of magnitude lower than that of metal and becomes very difficult to realize matching sealing. To achieve high quality of quartz glass -metal sealing, one method is to use a group of glass with similar thermal expansion coefficients as a transition zone to realize matching sealing; another method is to weld quartz glass and metal with solder. Cui Xing-qiang et al firstly metallized the surface of quartz glass [7,8], and then welded quartz glass and metal with solder. In this study, an active Sn-Ti alloy was developed to weld quartz glass and metal directly. To achieve hermetic seals, the key is to form a continuous transition interface between Sn-Ti alloy and quartz glass. Many literatures widely reported Sn-Ti alloy [9–16], but few reported it was applied to weld quartz glass and metal.

The purpose of the present work was to analyze the interface structure between Sn-Ti alloy and quartz glass, and verify the reliability of the seal.

2. Experimental procedure

The size of quartz glass used in the experiment is $\Phi 18 \times 1$mm, with surface roughness less than 0.5 nm and SiO_2 content of more than 99.99 wt%. Sn-Ti alloy is pressed into slice with a thickness of about 0.2 mm, and the Ti content is less than 6%. The sealing process of Sn-Ti alloy- quartz glass was as follows: Sn-Ti alloy slice was firstly cut into disk with the diameter of 18 mm, and then holden in two pieces of quartz glass; the sample was placed in the vacuum brazing furnace, and then heated up to 800 °C for 10 min when vacuum degree was not higher than 5×10^{-3} Pa; with the furnace cooling to room temperature, the sample was removed and cut to two pieces along the diameter direction.

The interfacial structure of the sample was investigated by transmission electron microscope (JEOL JEM-2010 UHR) operated at 300 kV. Specimens for TEM observation were prepared by FIB with a microsampling system (Hitachi FE-2100) using Ga ion beam.

3. Results and discussion

Fig. 1 is an overview of the interface morphology between Sn-Ti alloy and quartz glass. Fig. 1(a) is seen that Sn-Ti alloy is tightly attached to SiO_2 by a continuous interface layer with a thickness

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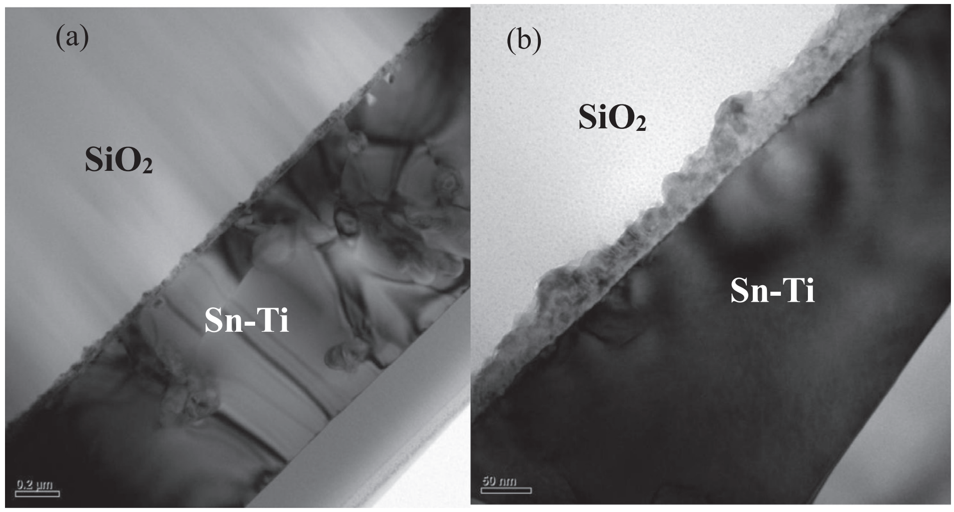

Fig. 1. Interface morphology between Sn-Ti alloy and quartz glass.

of several tens of nanometer, and no holes or cracks are observed in the interface. It shows that the interface structure has good stability and can withstand the thermal stress caused by the mismatch of the thermal expansion coefficients. Fig. 1(b) shows that the boundary between SiO_2 and the interface is rough, but the boundary between Sn-Ti alloy and the interface smooth. It suggests that mutual diffusion or reaction took place between Sn-Ti alloy and SiO_2, but the extent of mutual diffusion or reaction in different regions is not consistent.

Nano scale EDS mapping was carried out to further clarify the microstructure of the interface between Sn-Ti alloy and quartz glass, as shown in Fig. 2. Fig. 2(c) shows that Sn element exists only in Sn-Ti alloy. Fig. 2(d) shows that Ti element is enriched in the interlayer, but not found in SnTi alloy. Fig. 2(e) shows that Si element exists only in SiO_2 substrate and the interface. Fig. 2(f) shows that O content in the inter layer is the highest. So it is clearly seen that the interlayer is composed of three elements: Ti, Si and O, which is different from Sn-Ti alloy and quartz glass. And it can be furtherly divided into two parts: a thin layer of about 2 nm rich in Ti element and O element near Sn-Ti alloy and a thick layer of more than 10 nm rich in Ti element, Si element and O element near SiO_2. Combining this result with the next HRTEM results, it is believed that the interface consists of a TiO_2 layer and a Ti-Si-O layer, whose structure is completely different from that of Sn-Ti alloy and quartz glass.

HRTEM image of the different regions in the interface was shown in Fig. 3. The SAED image (as shown in Fig. 3(e)) of the interlayer shows a clear polycrystalline phase structure with no amorphous phase. Sn-Ti alloy is composed of Tetragonal Sn and a few series of SnTi intermetallic compounds. In the HRTEM images, a single big Sn grain can be observed, and it is connected to a 2-nm TIO_2 layer made of tiny nanocrystals, so the phase boundary is not straight. This microstructure increases the contact area between the TiO_2 layer and Sn grain, reduces the stress concentration of the phase boundary, and thus enhances the bonding strength. Adjacent to the TiO_2 layer, nanocrystalline TiO_2, $TiSi_2$ and Ti_5Si_3 are identified. These three phases are mixed together and then make up a multiphase layer. Fig. 1 shows that the multiphase layer has a coarser boundary with the amorphous quartz glass, so the bonding strength is high enough to prevent separation

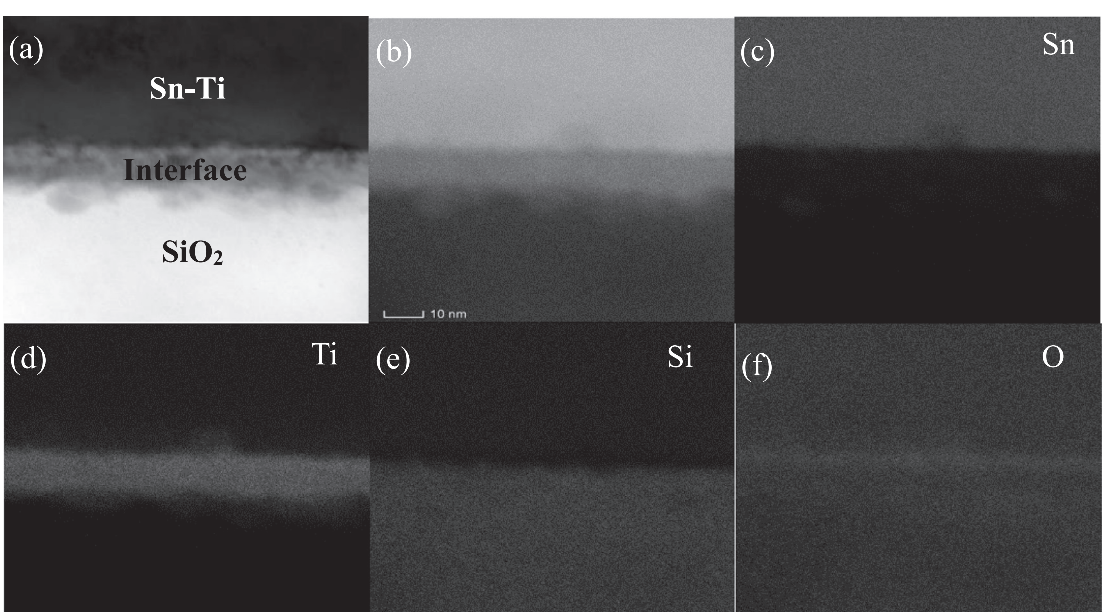

Fig. 2. EDS mapping of the interface.

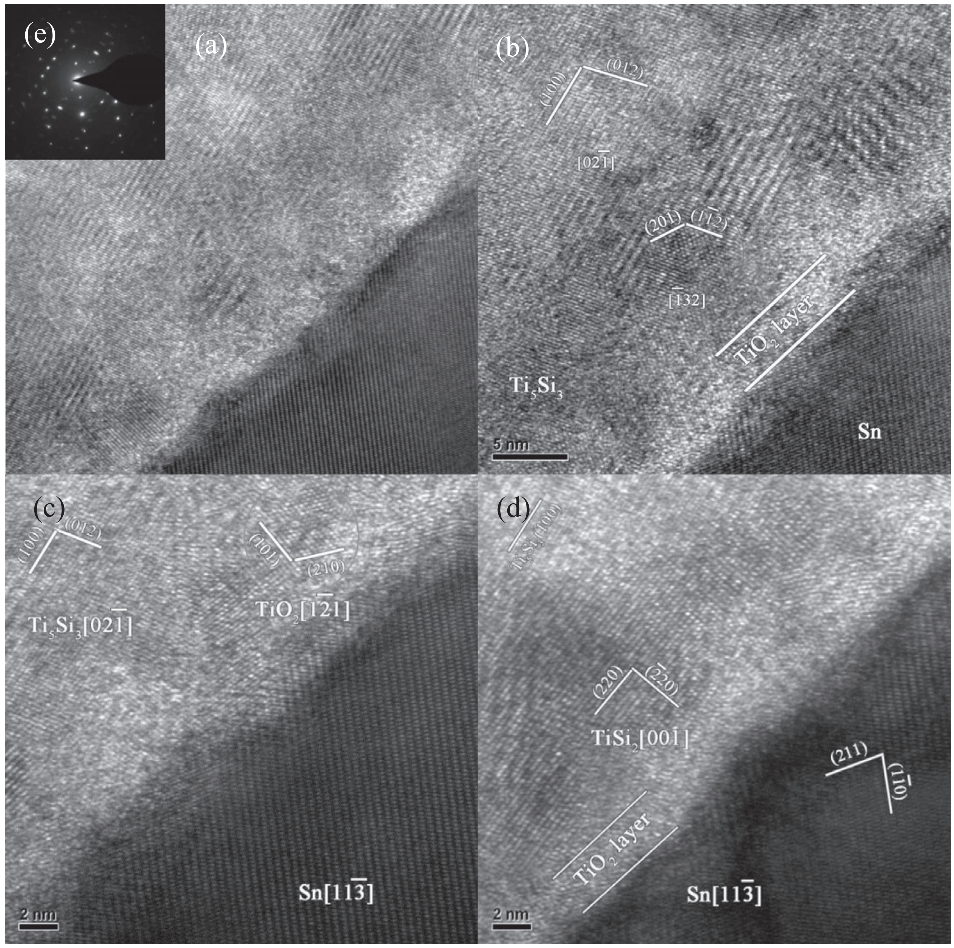

Fig. 3. HRTEM images of the different regions in the interface.

of Sn-Ti alloy and quartz glass. According to the above analysis results, the interlayer is the bond to achieve the tight connection between Sn-Ti alloy and quartz glass. This microstructure also blocks the diffusion path of gas molecules along the interface, then realizes hermetic seal between Sn-Ti alloy and quartz glass.

According to the above interfacial microstructure observation, the interface can be ascribed to the reaction of active Ti atoms and SiO_2. Ti- SiO_2 interface reaction has been thoroughly investigated in the past decades [17–25]. Various Ti-silicides and Ti-oxides are found as reaction products. In this experiment, Ti_5Si_3, $TiSi_2$, and TiO_2 are observed, and the interface consists of a TiO_2 layer and a Ti-Si-O multiphase layer. Combining these results with those reported, the reaction mechanism of the interface between Sn-Ti alloy and quartz glass can be deduced as follows:

At first, as the temperature rises to 800 °C, Sn-Ti alloy melts into fusing fluids. and Ti atoms are evenly distributed in the fusing fluids. The Sn-Ti melt spreads on the surface of the SiO_2 solid. At the initial stage of the reaction between Sn-Ti melt and SiO_2 solid, Ti atoms diffuse into the SiO_2 solid, and the O atoms into the Sn-Ti melt. In the direction of boundary to Sn-Ti melt, O atoms react with Ti atoms to form TiO (as formula (1)). A large number of Ti atoms diffusing from the boundary to SiO_2 solid react with Si atoms to form Ti_5Si_3 (as formula (2)) and O atoms diffusing from to Sn-Ti melt form TiO (as formula (1)).

$$\mathrm{Ti} + \mathrm{O} \rightarrow \mathrm{TiO} \tag{1}$$

$$5\mathrm{Ti} + 3\mathrm{Si} \rightarrow \mathrm{Ti_5Si_3} \tag{2}$$

With nanocrystalline TiO growing rapidly along the surface of SiO_2 solid and form a continuous TiO layer, Ti atoms with low concentration in the melt diffuse more and more slowly into SiO_2 solid due to the TiO layer block. It results in sharp decline in the amounts of Ti atoms reacted with SiO_2 until Ti atoms are exhausted. During this process, the diffusion rate of Ti atoms in different regions of SiO_2 solid surface becomes different because of the block effect of TiO layer. It leads to different Ti atoms concentration in different regions of SiO_2 solid and different thickness of the reaction interface. O atoms continue to spread into the TiO layer by interstitial diffusion mechanism and react with TiO to produce a compact and continuous nanocrystalline TiO_2 layer (as formula (3)). Ti_5Si_3 in some region further reacts with the excess Si atoms to form $TiSi_2$ (as formula (4)).

$$\mathrm{TiO} + \mathrm{O} \rightarrow \mathrm{TiO_2} \tag{3}$$

$$\mathrm{Ti_5Si_3} + 7\mathrm{Si} \rightarrow 5\mathrm{TiSi_2} \tag{4}$$

When the nanocrystalline TiO_2 layer at the solid-liquid interface is completely compact, the interfacial reaction becomes very slow and even stops with the rapid decrease of the diffusion rate of Ti atoms. Then a nano-layered-structure interface is born.

4. Conclusions

The results show that the interlayer is continuous and dense with a thickness of several tens of nanometer, and absent of holes and cracks. Interface region behaves a strong bonding strength, and can realize a reliable hermetic seal. The interface is characterized by a nano-layered-structure with two layers: one is the TiO_2 layer with a thickness of about 2 nm locates near Sn-Ti alloy, the other is polycrystalline multiphase layer near the quartz glass which is composed of $TiSi_2$, Ti_5Si_3 and TiO_2.

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